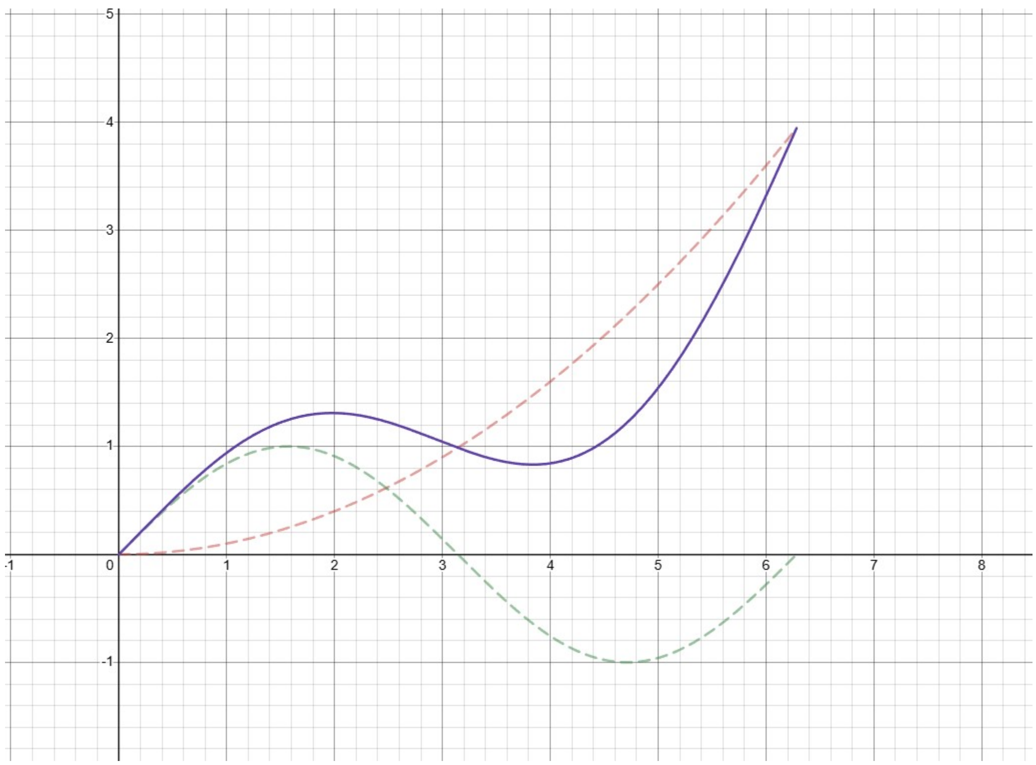
1 Addition of ordinates review

adding y-values for a given x-value

Consider graphing $f : (0, 2\pi) \to \mathbb{R}, f(x) = \frac{x^2}{10} + \sin(x)$



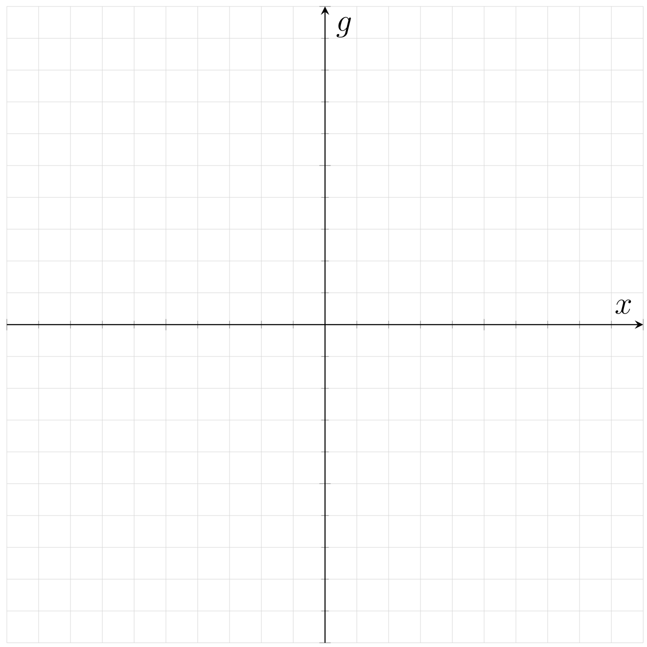
strategy?

- start with finding key points of each curve (if there are any)
- plot the curves individually
- add the height of each curve at discrete intervals (i.e at every tick)
- make sure the y values at key points match

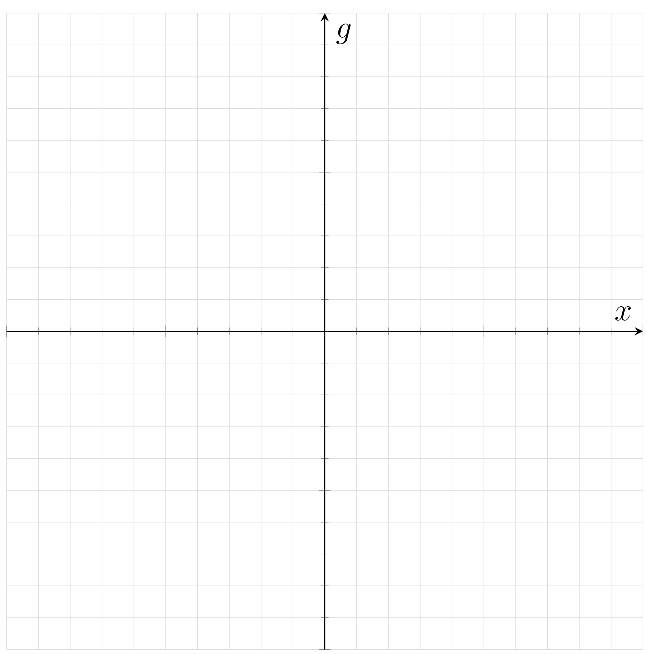
New way to think of dilation from the y axis! i.e : $f(x) = 2x^2 = x^2 + x^2$

2 Rational functions

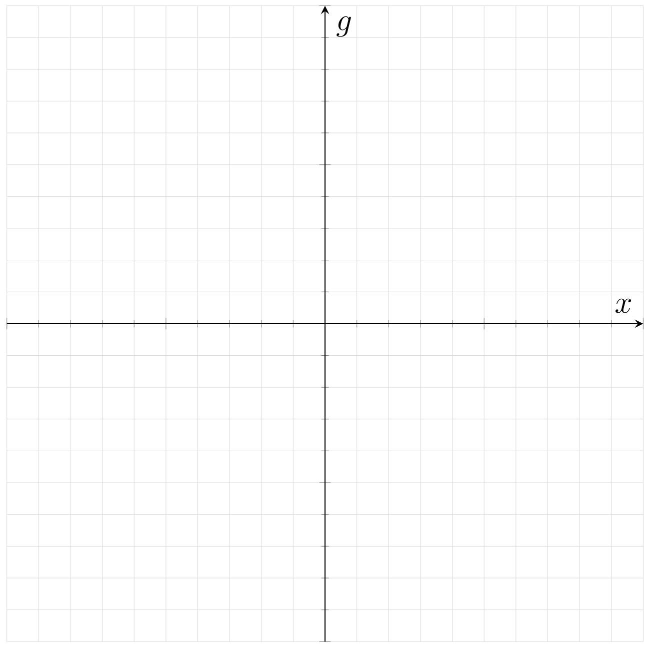
1. $g : \mathbb{R} \to \mathbb{R}, g(x) = \frac{1+x^2}{x}$



2. $g : \mathbb{R}^+ \to \mathbb{R}, g(x) = \frac{1+x^{\frac{5}{2}}}{x^2}$



3. (‡) $g : \mathbb{R} \to \mathbb{R}, g(x) = \frac{x^2|x| - 2x^2 + 1}{|x|}$



Trick with plotting graphs with modulus functions: Remember the effects of:

- $f(|x|)$
- $|f(x)|$

3 Asymptotes

Types of asymptotes:

- Vertical asymptotes : occur when the function approaches infinity as x approaches a certain value
  - often occur at values that **make the denominator zero**
- Horizontal asymptotes : occur when the function approaches a constant value as $x \to \infty$
- Oblique asymptotes : occur when the function approaches a linear function as $x \to \infty$

Which asymptotes can they cross?

- Vertical asymptotes : never, **because if denom = 0 then function is undefined**
- Horizontal asymptotes : can cross
- Oblique asymptotes : can cross

Try to find the asymptotes of the following functions by hand, then use mathematica to verify (plot + calculate limit if needed)

- $y = \frac{2x+3}{x-1}$
- $y = \frac{x}{x^2+1}$
- $y = \frac{2x^3+x^2+x}{x^2-9}$

4 Useful mathematica code

- Plotting: (This will be the most used function in both methods and spesh, memorise it!)

  ```
  Plot[<expr>, {x, <xmin>, <xmax>}, PlotRange -> {<ymin>,<ymax>}]
  ```

- Plotting multiple functions:

  ```
  Plot[{<expr1>, <expr2>, <expr3>}, {x, <xmin>, <xmax>}, PlotRange -> {<ymin>,<ymax>}]
  ```

- Calculating limits:

  ```
  Limit[<expr>, x -> <val>]
  ```